

MISC

Quadratic formula:

$$ax^2 + bx + c = 0 \Rightarrow x_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

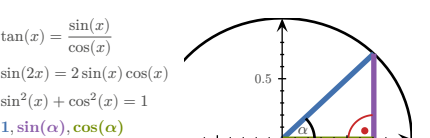
Monotonically in-/decreasing:

$$\forall x. (f'(x) > 0) \Rightarrow a > b \Leftrightarrow f(a) > f(b)$$
$$\forall x. (f'(x) < 0) \Rightarrow a > b \Leftrightarrow f(a) < f(b)$$

RSS:

$$RSS = \sum_i (y_i - f(x_i))^2$$

TRIGONOMETRY						
x	0	$30^\circ = \pi/6$	$45^\circ = \pi/4$	$60^\circ = \pi/3$	$90^\circ = \pi/2$	
$\sin(x)$	0	$1/2$	$\sqrt{2}/2$	$\sqrt{3}/2$	1	
$\cos(x)$	1	$\sqrt{3}/2$	$\sqrt{2}/2$	$1/2$	0	
$\tan(x)$	0	$1/\sqrt{3}$	1	$\sqrt{3}$		



VECTORS

Term	Definition
Norm/Length	$\ v\ = \sqrt{\sum_{i=1}^n v_i^2} = \sqrt{v^T v}$ $\nabla \ v\ = \frac{1}{\ v\ } v^T$ $\nabla \frac{1}{\ v\ } = -\frac{1}{\ v\ ^2} \nabla \ v\ = -\frac{1}{\ v\ ^3} v^T$ $\nabla \ r\ ^2 = 2 \ r\ \frac{1}{\ r\ } r^T = 2 r^T$
Distance	$d(u, v) = \ u - v\ $ $\nabla d(u, v) = \frac{1}{d(u, v)} ((u - v)^T (v - u)^T)$
Scalar product	$u \bullet v = u^T v = \sum_k u_k v_k$ $\nabla (u \bullet v) = (v^T \ u^T)$ $\nabla_u (u \bullet v) = v^T$

MATRICES

Inverting:

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \Rightarrow A^{-1} = \frac{1}{ad - cb} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

Transposing:

$$\begin{pmatrix} 1 & 0 \\ 0 & 2 \\ 3 & 1 \end{pmatrix}^T = \begin{pmatrix} 1 & 0 & 3 \\ 0 & 2 & 1 \end{pmatrix} \quad (A^T)^T = A \quad (A + B)^T = A^T + B^T$$
$$\begin{pmatrix} 1 \\ 4 \\ 5 \end{pmatrix}^T = (1 \ 4 \ 5) \quad (\lambda A)^T = \lambda A^T \quad (AB)^T = B^T A^T$$

Multiplication:

$$\begin{pmatrix} 2 & 3 & 1 \\ 4 & -1 & 7 \end{pmatrix} \cdot \begin{pmatrix} 6 & 4 \\ 1 & 0 \\ 8 & 9 \end{pmatrix} = \begin{pmatrix} (2 \cdot 6 + 3 \cdot 1 + 1 \cdot 8) & (2 \cdot 4 + 3 \cdot 0 + 1 \cdot 9) \\ (4 \cdot 6 + -1 \cdot 1 + 7 \cdot 8) & (4 \cdot 4 + -1 \cdot 0 + 7 \cdot 9) \end{pmatrix} = \begin{pmatrix} 23 & 17 \\ 79 & 79 \end{pmatrix}$$

PROPERTIES

$A \cdot A^{-1} = 1$ $\ker A = \{v \mid Av = 0\}$

$A + B = B + A$ $A \cdot B \neq B \cdot A$

$A + (B + C) = (A + B) + C$ $A(BC) = (AB)C$

$A(B + C) = AB + AC$ $(A + B)C = AC + BC$

DETERMINANT

$\det(a) = a$ $\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - cb$

$$\det \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} = a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{31}a_{22}a_{13} - a_{32}a_{23}a_{11} - a_{33}a_{21}a_{12}$$
$$\det(A \cdot B) = \det(A) \cdot \det(B) \quad \det(1) = 1$$
$$\det(A^{-1}) = \frac{1}{\det(A)} \quad \det(A^T) = \det(A)$$

$\det \begin{pmatrix} 0 & 2 & 1 \\ 1 & 0 & 1 \\ 3 & 4 & 2 \end{pmatrix} = -2 \det \begin{pmatrix} 1 & 1 \\ 3 & 2 \end{pmatrix} + 0 \det \begin{pmatrix} 0 & 1 \\ 3 & 2 \end{pmatrix} - 4 \det \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$
$$= -2(2 - 3) + 0 - 4(-1) = 2 + 4 = 6$$

Schema :

$$\begin{pmatrix} + & - & + & \dots \\ - & + & - & \dots \\ + & - & + & \dots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}$$

EIGENVALUES

$M \in \mathbb{R}^{n \times n}, \quad M^T = M$

$$\det(M) = \lambda_1 \cdot \lambda_2 \cdot \dots \cdot \lambda_n$$

Eigenvector v for the eigenvalue λ of M :

$$Mv = \lambda v$$
$$\Leftrightarrow (M - \lambda 1)v = 0$$

Eigenvalue λ of M :

$$\det(M - \lambda 1) = 0$$

Finding eigenvalues:

$$A = \begin{pmatrix} 1 & 1 \\ -2 & 4 \end{pmatrix}$$
$$\det(A - \lambda 1) = \det \begin{pmatrix} 1 - \lambda & 1 \\ -2 & 4 - \lambda \end{pmatrix} = (1 - \lambda)(4 - \lambda) - (-2) \cdot 1 = \lambda^2 - 5\lambda + 6 = (\lambda - 3)(\lambda - 2)$$

or alternatively

$$m = \frac{1}{2} \operatorname{tr} \begin{pmatrix} 1 & 1 \\ -2 & 4 \end{pmatrix} = \frac{1 + 4}{2} = \frac{\lambda_1 + \lambda_2}{2}$$
$$p = \det \begin{pmatrix} 1 & 1 \\ -2 & 4 \end{pmatrix} = 4 + 2 = \lambda_1 \lambda_2$$
$$\lambda_{1,2} = m \pm \sqrt{m^2 - p} = \frac{5}{2} \pm \sqrt{\frac{25}{4} - 6} = \frac{5}{2} \pm \frac{1}{2}$$

Indices

$$A \in \mathbb{R}^{n \times m}, B \in \mathbb{R}^{m \times r}, \quad A \cdot B \in \mathbb{R}^{n \times r}$$
$$(A \cdot B)_{ij} = \sum_{k=1}^m A_{ik} B_{kj} = \sum_k A_{ik} B_{kj}$$
$$((A \cdot B)^T)_{ij} = (A \cdot B)_{ji} = \sum_k A_{jk} B_{ki} = \sum_k A_{jk} B_{ki}$$
$$(A \cdot B^T)_{ij} = \sum_k A_{ik} B_{kj}^T = \sum_k A_{ik} B_{jk}$$
$$a^T \cdot b \cdot b = \sum_{ij} (a^T)_i B_{ij} b_j = \sum_{ij} a_i B_{ij} b_j$$
$$(B \cdot a)(C \cdot a) = \sum_i (B \cdot a)_i (C \cdot a)_i$$

INDICES

Addition/Subtraction

$$A \in \mathbb{R}^{n \times m}, B \in \mathbb{R}^{n \times m}$$
$$(A + B)_{ij} = A_{ij} + B_{ij}$$
$$(A - B)_{ij} = A_{ij} - B_{ij}$$

KRONECKER DELTA

$\delta_{ij} = (\tilde{e}_i)_j = \begin{cases} 1, & i = j \\ 0, & \text{otherwise} \end{cases}$

$$(E_{ij})_{kr} = \delta_{ik} \delta_{jr}$$
$$(E_{rst})_{ijk} = \delta_{ri} \delta_{sj} \delta_{tk}$$

BASIS VECTORS

$\mathbb{R}^3$

$$e_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, e_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, e_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

UNIT MATRICES

$\mathbb{R}^2$

$$E_{11} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, E_{12} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, E_{21} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, E_{22} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

AFFINE TRANSFORMATIONS

$M \in \mathbb{R}^{m \times n}$

$$A_{b,M} : \begin{cases} \mathbb{R}^m \rightarrow \mathbb{R}^n \\ x \mapsto b + M \cdot x \end{cases}$$
$$L_A(\vec{0}) = \vec{0}$$

DERIVATIVES

1	$\rightarrow 0$	x	$\rightarrow 1$
x^2	$\rightarrow 2x$	x^n	$\rightarrow nx^{n-1}$
$\frac{1}{x}$	$\rightarrow -\frac{1}{x^2}$	$\sqrt{x}$	$\rightarrow \frac{1}{2\sqrt{x}}$
e^x	$\rightarrow e^x$	e^{-x}	$\rightarrow -e^{-x}$
$\ln(x)$	$\rightarrow \frac{1}{x}, x > 0$	$\ln(y \cdot x)$	$\rightarrow \frac{1}{x}, x > 0$
a^x	$\rightarrow \ln(a) \cdot a^x, a > 0, a \neq 1$	$\log_b(x)$	$\rightarrow \frac{1}{\ln(b) \cdot x}$
$\sin(x)$	$\rightarrow \cos(x)$	$\sin(2x)$	$\rightarrow 2 \cos(2x)$
$\cos(x)$	$\rightarrow -\sin(x)$	$\cos(ax)$	$\rightarrow -a \sin(ax)$
$\tan(x)$	$\rightarrow 1 + \tan^2(x) = \frac{1}{\cos^2(x)}$	$\operatorname{arccot}(x)$	$\rightarrow -\frac{1}{1 + x^2}$
$\cot(x)$	$\rightarrow -\frac{1}{\sin^2(x)}$	$\arcsin(x)$	$\rightarrow \frac{1}{\sqrt{1 - x^2}}$
$\arccos(x)$	$\rightarrow -\frac{1}{\sqrt{1 - x^2}}$	$\arctan(x)$	$\rightarrow \frac{1}{1 + x^2}$

Case	Rule
Addition	$(f(x) + g(x))' = f'(x) + g'(x)$
Multiplication	$(\alpha \cdot f(x))' = \alpha \cdot f'(x)$
Product rule	$(f \cdot g)' = f' \cdot g + f \cdot g'$ $(f \cdot g \cdot h)' = f' \cdot g \cdot h + f \cdot g' \cdot h + f \cdot g \cdot h'$
Chain rule	$(f(g(x)))' = f'(g(x)) \cdot g'(x)$
Quotient rule	$\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$

INTEGRALS

$\frac{1}{x}$	$\rightarrow \ln(x) + c$	$\ln(x)$	$\rightarrow x \ln(x) - x + c$
e^x	$\rightarrow e^x + c$	$\frac{1}{\sqrt{1 - x^2}}$	$\rightarrow \arcsin(x) + c$
$\sin(x)$	$\rightarrow -\cos(x) + c$	$-\frac{1}{\sqrt{1 - x^2}}$	$\rightarrow \arccos(x) + c$
$\cos(x)$	$\rightarrow \sin(x) + c$	$\frac{1}{1 + x^2}$	$\rightarrow \arctan(x) + c$
$\tan(x)$	$\rightarrow -\ln(\cos(x)) + c$	$\frac{1}{a + x^2}$	$\rightarrow \frac{1}{\ln(a^2)} a^x + c$
x^a	$\rightarrow \frac{1}{a + 1} x^{a+1} + c, a \neq -1$	a^{bx}	$\rightarrow \frac{1}{\ln(a^b)} a^x + c$
a^x	$\rightarrow \frac{1}{\ln(a)} a^x + c$		
a^x	$\rightarrow e^{\ln(a) \cdot x}$		

$\arctan(x) \rightarrow x \arctan(x) - \frac{1}{2} \ln(|1 + x^2|) + c$

$\arccos(x) \rightarrow x \arccos(x) - \sqrt{1 - x^2} + c$

$\arcsin(x) \rightarrow x \arcsin(x) + \sqrt{1 - x^2} + c$

RULES

$\int f(g(x))g'(x) dx = F(g(x)) + \text{const}$

$$\int f(x)g(x) dx = F(x)g(x) - \int F(x)g'(x) dx$$

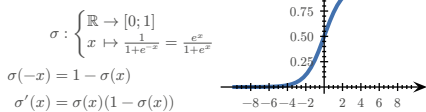
FUNCTIONS OF SEVERAL VARIABLES

Term	Definition
Real valued function	$R \subset \mathbb{R}$
Vector valued function	$R \subset \mathbb{R}^m$
A function of n variables	$D \subset \mathbb{R}^n$
n -dimensional curve	$c : \mathbb{R} \rightarrow \mathbb{R}^n$
n -dimensional hyper-surface	$s : \mathbb{R}^n \rightarrow \mathbb{R}$
Reparametrization	$c(t) \rightarrow d(s) = c(h(s))$

Softmax function:

$$\sigma : \begin{cases} \mathbb{R}^n \rightarrow [0, 1]^n \\ x_i \mapsto e^{x_i} \left(\sum_{j=1}^n e^{x_j} \right)^{-1} \end{cases}$$
$$J_{ij} \sigma(x) = \delta_{ij} \frac{e^{x_i}}{\sum_{k=1}^n e^{x_k}} - \frac{e^{x_i} e^{x_j}}{\left(\sum_{k=1}^n e^{x_k} \right)^2}$$

Sigmoid function:



DERIVATIVES

Function	Derivative
$f : \mathbb{R} \rightarrow \mathbb{R}^n$ $x \mapsto y$	$f' : \mathbb{R} \rightarrow \mathbb{R}^n = \begin{pmatrix} \frac{d}{dx} f_1(x) \\ \vdots \\ \frac{d}{dx} f_n(x) \end{pmatrix}$
$f : \mathbb{R}^n \rightarrow \mathbb{R}$ $x \mapsto y$	$f' : \mathbb{R}^n \rightarrow \mathbb{R}^{1 \times n} = \nabla f$ $= \left(\frac{\partial}{\partial x_1} f(x), \frac{\partial}{\partial x_2} f(x), \dots, \frac{\partial}{\partial x_n} f(x) \right)$
$f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ $x \mapsto y$	$f' : \mathbb{R}^n \rightarrow \mathbb{R}^{m \times n} = J_f = \frac{\partial f}{\partial x} \Big _{x=x_0}$ $= \begin{pmatrix} \nabla f_1(x) \\ \vdots \\ \nabla f_m(x) \end{pmatrix} = \begin{pmatrix} \frac{\partial}{\partial x_1} f_1(x) & \dots & \frac{\partial}{\partial x_n} f_1(x) \\ \vdots & \ddots & \vdots \\ \frac{\partial}{\partial x_1} f_m(x) & \dots & \frac{\partial}{\partial x_n} f_m(x) \end{pmatrix}$

PARTIAL VS TOTAL DERIVATIVE

$f(x, y) = x^2 + 2y$

$$\frac{\partial f}{\partial x} = 2x$$
$$\frac{df}{dx} = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} \cdot \frac{dy}{dx} = 2x + 2 \cdot \frac{dy}{dx} \mid (y \text{ depend on } x)$$

GRADIENT VS TOTAL DERIVATIVE

$f(x, y) = xy e^y$

$$x(t) = e^t$$
$$y(t) = t^2$$
$$\nabla f = (y e^y, x e^y + x y e^y) = e^y (y, x + x y)$$
$$\frac{d}{dt} f = e^y (y, x + x y) \cdot \frac{d}{dt} \begin{pmatrix} e^t \\ t^2 \end{pmatrix} = \frac{d}{dt} (e^t t^2 e^t) = (2t + t^2 + 2t^3) e^{t^2 + t}$$

GRADIENTS

Common gradients:

$$f : \mathbb{R}^n \rightarrow \mathbb{R}, x \in \mathbb{R}^n, \quad a \in \mathbb{R}^n, c \in \mathbb{R}, A \in \mathbb{R}^{n \times n}$$
$$f(x) = \|x\|^2 \Rightarrow \nabla f = 2x^T$$
$$f(x) = a^T x + c \Rightarrow \nabla f = a^T$$
$$f(x) = x^T A x \Rightarrow \nabla f = x^T (A + A^T)$$

$f(x) = \frac{1}{2} x^T A x + b^T x + c \Rightarrow \nabla f = x^T \frac{A + A^T}{2} + b^T = x^T A + b^T \text{ if } A^T = A$

$g : \mathbb{R}^m \rightarrow \mathbb{R}, B \in \mathbb{R}^{m \times n}$

$$f(x) = g(Bx + b) \Rightarrow \nabla f = (\nabla g(Bx + b))^T B$$

JACOBIAN MATRIX

$f : \mathbb{R}^n \rightarrow \mathbb{R}^m, \quad x_0 \in \mathbb{R}^n$

$$J_f(x_0) : \mathbb{R}^{m \times n} \quad J_f(x_0)_{ij} = \frac{\partial}{\partial x_j} f_i(x_0)$$
$$f : \mathbb{R}^2 \rightarrow \mathbb{R}^2 \quad J_f(x, y) = \begin{pmatrix} \frac{\partial x_1}{\partial x_2} & \frac{\partial y_1}{\partial y_2} \end{pmatrix}$$

Euclid₂ $(\mathbb{R}^n, x_0)$ $\xrightarrow{g}$ $(\mathbb{R}^m, g(x_0))$ $\xrightarrow{f}$ $(\mathbb{R}^k, f(g(x_0)))$

Mat_R $\mathbb{R}^n$ $\xrightarrow{J_g(x_0)}$ $\mathbb{R}^m$ $\xrightarrow{J_f(g(x_0))}$ $\mathbb{R}^k$

$\xRightarrow{D} \xRightarrow{f \circ g}$

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Chain rule:

$f : \mathbb{R}^n \rightarrow \mathbb{R}^m, g : \mathbb{R}^m \rightarrow \mathbb{R}^k, h : \mathbb{R}^n \rightarrow \mathbb{R}^k = g \circ f$

$$J_h = J_g(f(x)) \cdot J_f(x)$$

Common jacobians:

$f(x) = a + M \cdot x \Rightarrow J_f = M$

$f(x) = x \Rightarrow J_f = 1$

$f(x) = (x^T x) x \Rightarrow J_f = 2x x^T + (x^T x) \cdot 1 \Rightarrow (J_f)_{ij} = 2x_j x_i + \sum_k x_k^2 \delta_{ij}$

$f(x) = \|x\|^2 \Rightarrow J_f = (2x)^T$

$f_i : \mathbb{R}^n \rightarrow \mathbb{R}^n \quad x \mapsto a_i + M_i x$

$g(x) = f_n(f_{n-1}(\dots(f_1(x)))) \Rightarrow J_g = M_n \cdot M_{n-1} \cdot \dots \cdot M_1$

LINEARISATION

$f : \mathbb{R}^n \rightarrow \mathbb{R}^m, x_0 \in \mathbb{R}^n$

$$L_f(x) = f(x_0) + J_f(x_0) \cdot (x - x_0)$$

For finding better approximations for x_0 :

- 1) Calculate J_f , then $J_f(x_0)$, then L_f
- 2) Find x so that $L_f(x) = 0$ (Gauss)

For calculating an approximate value at x :

- 1) x_0 = round x to ints
- 2) Calculate $L_f(x)$ at x_0

SECOND DERIVATIVE

$$\frac{\partial}{\partial x} \left(\frac{\partial}{\partial x} (f(x, y)) \right) = \frac{\partial^2 f(x, y)}{\partial x^2} = \partial_{xx} f(x, y)$$

HESSIAN MATRIX

$f : \mathbb{R}^n \rightarrow \mathbb{R}, \quad x \in \mathbb{R}^n, \quad H(x) \in \mathbb{R}^{n \times n}$

$$H(x, y) = \begin{pmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{pmatrix} = J(\nabla f)$$
$$(H(x))_{ij} = \frac{\partial}{\partial x_i} \frac{\partial}{\partial x_j} f(x)$$
$$\partial_i \partial_j f = \partial_j \partial_i f$$

Common Hessians:

$f(x) = \frac{1}{2} x^T A x + b^T x + c \Rightarrow H_f = \frac{A + A^T}{2}$

$f(x) = g(Ax + b) \Rightarrow H_f = A^T H_g(Ax + b) A$

$g : \mathbb{R}^m \rightarrow \mathbb{R}$

EXTREMAL POINTS

Quadratic form:

$$M \in \mathbb{R}^{n \times n} \qquad q_M(v) = v^T M v \qquad v \in \mathbb{R}^n$$
$$H = \frac{1}{2}(M + M^T) \quad \forall v \in \mathbb{R}^n. \quad (q_M(v) = q_H(v))$$

$\forall v. (v^T H v > 0) \vee \forall \lambda. (\lambda > 0) \Rightarrow c_0$ is a **local minimum**, H is **positive definite**

$\forall v. (v^T H v \geq 0) \vee \forall \lambda. (\lambda \geq 0) \Rightarrow c_0$ is a **local minimum** or **non-extremal**, H is **positive semi-definite**

$\forall v. (v^T H v < 0) \vee \forall \lambda. (\lambda < 0) \Rightarrow c_0$ is a **local maximum**, H is **negative definite**

$\forall v. (v^T H v \leq 0) \vee \forall \lambda. (\lambda \leq 0) \Rightarrow c_0$ is a **local maximum** or **non-extremal**, H is **negative semi-definite**

$\forall v. (v^T H v \neq 0) \vee \forall \lambda. (\lambda \neq 0) \Rightarrow c_0$ is a **saddle point**, H is **indefinite**

Finding stationary points:

1) Calculate ∇f , then H_f
2) Find stationary point(s) x so that $\nabla f(x) \stackrel{!}{=} 0$
3) Find type of point using technique of completing squares on $H_f(x)$

$$(v_1, v_2, v_3) \cdot \begin{pmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{pmatrix} \cdot \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix}$$
$$= v_1^2 h_{11} + v_2^2 h_{22} + v_3^2 h_{33} + 2v_1 v_2 h_{12} + 2v_2 v_3 h_{23} + 2v_1 v_3 h_{13}$$

Technique of completing squares:

$$\frac{x^2 - 4xy + 1}{\sqrt{\quad}} \xrightarrow{+2x} (x - 2y)^2 - (-2y)^2 + 1$$
$$= (x - 2y)^2 - 4y^2 + 1$$

HYPER-SURFACES

Chain rule:

$f: \mathbb{R}^n \rightarrow \mathbb{R}^m, g: \mathbb{R}^m \rightarrow \mathbb{R}, h: \mathbb{R}^n \rightarrow \mathbb{R} = g \circ f$
$$\nabla h(x) = \nabla g(f(x)) = \nabla g(f)|_{f=f(x)} \cdot \frac{\partial}{\partial x} f(x)$$

Differentiation properties:

$f: \mathbb{R}^n \rightarrow \mathbb{R}, g: \mathbb{R}^n \rightarrow \mathbb{R}$
$$\nabla(f + g) = \nabla f + \nabla g$$
$$\nabla(f - g) = \nabla f - \nabla g$$
$$\nabla(f \cdot g) = g \nabla f + f \nabla g$$
$$\nabla \frac{f}{g} = \frac{g \nabla f - f \nabla g}{g^2}$$

COMMUTATIVE DIAGRAMS

$f: D \rightarrow M$
 $g: N \rightarrow R$
 $h: D \rightarrow R$
 $= g \circ f$

$$D \xrightarrow{f} M \xrightarrow{g} R$$
$$\quad \searrow \quad \quad \nearrow$$
$$\quad \quad \quad h$$

GRADIENT DESCENT / NEWTON'S METHOD

$GD \quad x_{i+1} = x_i - \gamma \cdot \nabla f(x_i)$
 $N \quad x_{i+1} = x_i - \gamma \cdot (H_{f(x_i)})^{-1} \cdot \nabla f(x_i)$
 γ = step size
termination criterion $\varepsilon > |x_{i+1} - x_i|$

Gradient descent doesn't always produce global minimum:

– If the function has more than one local minimum it could output that $\rightarrow$ run with different starting points
– Could output a stationary point that is not minimal (unlikely). $\rightarrow$ run with different starting points
– Could not terminate at all if step size / learning rate is so big (overshoots minimal point). Likely to happen if the graph of the function is narrow around the minimal point. $\rightarrow$ choose smaller step size

LOGISTIC REGRESSION

$\mathbb{R}^{128 \times 256 \times 3} \xrightarrow{b(x)} \mathbb{R} \xrightarrow{\sigma(m \cdot b - q)} [0; 1]$
 $\searrow \text{cat}(x)$
 $\quad \quad \quad \downarrow \mathbb{1}_{(t; \infty)}(\sigma)$
 $\quad \quad \quad \{0; 1\}$

$b(x)$ = Feature function
 $\sigma(m \cdot b - q)$ = Probability function
 m = Weight
 q = Bias
 $\mathbb{1}_{(t; \infty)}(\sigma)$ = Indicator function
 t = Threshold

PROBABILITY THEORY

Term	Definition
Distributed	$\sim$
Normal distribution	$\mathcal{N}(\mu, \sigma)$
Uniform distribution	$\text{unif}(a, b)$
Probability measure	$\mathbb{P}(E)$
Density	$f_\theta(x_i)$
Likelihood function	$L(\theta) = \prod_{i=1}^n f_\theta(x_i)$ e.g. $L(\theta) = \prod_{x \in S_{Y_{\text{obs}}}} p_\theta(f(x)) \prod_{x \in S_{Y_{\text{un}}}} 1 - p_\theta(f(x))$
Log-likelihood function	$\ln(L(\theta)) = \sum_{i=1}^n \ln(f_\theta(x_i))$
Cost function (negative log-likelihood)	$c(\theta) = -\ln(L(\theta)) = -\sum_{i=1}^n \ln(f_\theta(x_i))$
Indicator function	$\mathbb{1}_{(a; b)}(x) = \begin{cases} 1 & x \in (a; b) \\ 0 & x \notin (a; b) \end{cases}$
CDF	Cumulative Distribution Function F
PDF	Probability Density Function f

KOLMOGOROV AXIOMS

$\mathbb{P}(\Omega) = 1$
 $\mathbb{P}(\varnothing) = 0$
 $\forall A, B \subset \Omega, A \cap B = \varnothing. (\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B))$
 $\forall A, B \subset \Omega. (\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B))$

STATISTICALLY INDEPENDENT

$X: \Omega \rightarrow \mathbb{R}$ and $Y: \Omega \rightarrow \mathbb{R}$ are **statistically independent**, if the probability density f_Y of the random variable

$$V: \begin{cases} \Omega \rightarrow \mathbb{R}^2 \\ \omega \mapsto \begin{pmatrix} X(\omega) \\ Y(\omega) \end{pmatrix} \end{cases}$$

satisfies

$$f_V(x, y) = f_X(x) \cdot f_Y(y)$$

RELATING RANDOM VARIABLES

$$\begin{array}{ccccc} \Omega & \xrightarrow{X} & \mathbb{R} & \xrightarrow{F_X} & [0; 1] \\ & \searrow & \uparrow g & \nearrow & \\ & & \mathbb{R} & & \end{array}$$
$$F_X(g(y)) = P(X(\omega) \leq g(y)) = P(g(Y(\omega)) \leq g(y)) \Rightarrow f_X(g(y)) \cdot g'(y) = f_Y(y)$$

(by differentiation)

$$g(Y(\omega)) = X(\omega) \Rightarrow F_X(g(y)) = P(X(\omega) \leq g(y)) = P(g(Y(\omega)) \leq g(y))$$
$$\Rightarrow \begin{cases} F_X(g(y)) = P(Y(\omega) \leq y) = F_Y(y) & g \text{ strictly increasing} \\ F_X(g(y)) = P(Y(\omega) \geq y) = 1 - F_Y(y) & g \text{ strictly decreasing} \end{cases}$$
$$|f_Y(y) dy| = |f_X(x) dx| \quad \text{if monotonically in/decreasing}$$

MAXIMUM LIKELIHOOD PRINCIPLE

= Finding optimal parameters θ for fitting distribution to data

1) Calculate cost function c with given distribution function f_θ , parameters θ and data points $x_1, x_2, \dots, x_n$
2) Calculate stationary point(s) of c (set ∇c to 0)
 $\Rightarrow$ Find optimum (gradient descent/newton's method if calculation not possible)

DISCRETE PROBABILITY DISTRIBUTION

Let $\Omega = \{e_1, e_2, \dots, e_n\}$ be enumerable

$\mathbb{P}(\{e_k\}) = p_k, \quad k \in \{1, 2, \dots, n\}, \quad E \subset \Omega$
$$\mathbb{P}(E) = \sum_{e \in E} p(e)$$
$$F(x) = \sum_{e \in \Omega, e \leq x} p(e)$$

RULES

$\forall e \in \Omega. (0 \leq p(e) \leq 1)$
$$\sum_{e \in \Omega} p(e) = 1$$
$$\lim_{x \rightarrow -\infty} F(x) = 0$$
$$0 \leq F(x) \leq 1$$
$$\int_{-\infty}^{\infty} f(t) dt = 1$$
$$\lim_{t \rightarrow \infty} F(x) = 1$$
$$f(t) \geq 0 \text{ for } t \in \mathbb{R}$$

$F(x)$ is monotonically increasing & continuous from the right

CDF

$\text{unif}(0, 1)$

$F(x)$

PDF

$\text{unif}(0, 1)$

$f(x)$

UNIVARIATE UNIFORM DISTRIBUTION

$X \sim \text{unif}(a, b), \quad \Omega \subset \mathbb{R}$
$$f_X(x) = \begin{cases} \frac{1}{b-a} & \text{if } x \in (a; b) \\ 0 & \text{else} \end{cases} = \mathbb{1}_{(a; b)}(x) \cdot \frac{1}{b-a} = F'_X$$
$$F_X(x) = \mathbb{P}((-\infty; x] \cap \Omega) = \mathbb{1}_{[a; b]}(x) \cdot \frac{x-a}{b-a} + \mathbb{1}_{(b; \infty)}(x)$$
$$= \mathbb{P}(X \leq x) = \int_{-\infty}^x f_X(t) dt$$

UNIVARIATE NORMAL DISTRIBUTION

Standard normal distribution:

$$\varphi(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}$$

General normal distribution:

$$f(x|\mu, \sigma) = \frac{1}{\sigma} \varphi\left(\frac{x-\mu}{\sigma}\right)$$
$$x \sim \mathcal{N}(\mu, \sigma)$$
$$f(x|\mu, \sigma) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(x-\mu)^2}$$
$$\mu \approx \bar{x} = \frac{1}{N} \sum_{i=1}^N x_i$$
$$\sigma^2 \approx \text{var} = \frac{1}{N-1} \sum_{i=1}^N (x_i - \bar{x})^2$$

Univariate normal distribution:

$$\text{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt$$

Error function:

$$\text{CDF}$$
$$\text{PDF}$$

MULTIVARIATE NORMAL DISTRIBUTION

Σ = covariance matrix, $V: \Omega \rightarrow \mathbb{R}^n$
 $V \sim \mathcal{N}(\mu, \Sigma)$
$$f_V(v) = \frac{1}{\sqrt{(2\pi)^n \det \Sigma}} \cdot e^{\frac{1}{2} v^T \Sigma^{-1} (v - \mu)}$$
$$q_{\Sigma^{-1}}(v - \mu) = -\frac{1}{2} (v - \mu)^T \Sigma^{-1} (v - \mu)$$

EXAMPLES

WORKING WITH INDICES

$x, y \in \mathbb{R}^n, f: \mathbb{R}^n \rightarrow \mathbb{R}$
$$f(x) = (x - y)^T (3x - y) = \sum_{i=1}^n (x_i - y_i)(3x_i - y_i)$$
$$\partial_{x_i} f(x) = (3x_i - y_i) + 3(x_i - y_i)$$
$$f(x) = \ln(x_1 \cdot x_2 \cdot \dots \cdot x_n) - |x|^2 = \sum_{i=1}^n (\ln(x_i) - x_i^2)$$
$$\partial_{x_i} f(x) = \frac{1}{x_i} - 2x_i$$
$$f(x) = \ln(y^T x) = \ln\left(\sum_{i=1}^n y_i x_i\right) \Rightarrow \nabla f = \frac{1}{y^T x} y^T$$

General normal distribution:

$$f(x|\mu, \sigma) = \frac{1}{\sigma} \varphi\left(\frac{x-\mu}{\sigma}\right)$$
$$x \sim \mathcal{N}(\mu, \sigma)$$
$$f(x|\mu, \sigma) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(x-\mu)^2}$$
$$\mu \approx \bar{x} = \frac{1}{N} \sum_{i=1}^N x_i$$
$$\sigma^2 \approx \text{var} = \frac{1}{N-1} \sum_{i=1}^N (x_i - \bar{x})^2$$

Univariate normal distribution:

$$\text{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt$$

CALCULATING LIKELIHOOD FUNCTION

Given the following data and the probability density of the form

$$f(t) = \lambda e^{-\lambda t} \cdot \mathbb{1}_{(0; \infty)}(t)$$

find a formula for the likelihood function:

$$l(\lambda) = f(7) \cdot f(16) \cdot f(32) \cdot f(44) = \lambda^4 e^{-99\lambda}$$
$$-\ln(l(\lambda)) = -\ln(\lambda^4 e^{-99\lambda}) = 99\lambda - 4 \ln(\lambda)$$

CALCULATING PROBABILITY DENSITY FUNCTION

EXPONENTIAL DISTRIBUTION

Given σ , Y and X , an exponentially distributed random var

$$f_X(x) = X_{[0; \infty)}(x) \cdot e^{-x}$$
$$Y = \sigma(X)$$
$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

calculate the probability density function $f_Y(y)$

$|f_Y(y) dy| = |f_X(x) dx|$
 $\stackrel{y>0}{\Leftrightarrow} f_Y(y) = f_X(x) \cdot \frac{dx}{dy}$
$$y = \frac{1}{1 + e^{-x}}$$
$$\Leftrightarrow x = -\ln\left(\frac{1}{y} - 1\right), \quad 0 < y < 1$$
$$\Rightarrow \frac{dx}{dy} = -\frac{1}{y} \cdot \frac{-1}{y^2} = \frac{1}{y - y^2}$$
$$\Rightarrow f_Y(y) = X_{(0; \infty)}(x) \cdot e^{-x} \cdot \frac{1}{y - y^2}$$
$$= \frac{1}{y^2} \cdot X_{(0; \infty)}\left(-\ln\left(\frac{1}{y} - 1\right)\right)$$
$$-\ln\left(\frac{1}{y} - 1\right) \geq 0$$
$$\stackrel{y>0}{\Leftrightarrow} y \geq \frac{1}{2}$$
$$\Rightarrow X_{[0; \infty)}\left(-\ln\left(\frac{1}{y} - 1\right)\right) \stackrel{y \in (0; 1)}{=} X_{\left[\frac{1}{2}; 1\right)}(y)$$
$$\Rightarrow f_Y(y) = \frac{1}{y^2} X_{\left[\frac{1}{2}; 1\right)}(y)$$

UNIFORM DISTRIBUTION

Given $X \sim \text{unif}(0, 1)$, $p > 0$, $Y = X^p$ calculate the probability density function $f_Y(y)$

$$F_X(x) = \mathbb{P}(X \leq x)$$
$$F_Y(y) = \mathbb{P}(Y \leq y) = \mathbb{P}(X^p \leq y) = \mathbb{P}\left(X \leq y^{\frac{1}{p}}\right) = F_X\left(y^{\frac{1}{p}}\right) = \mathbb{1}_{[0; 1]}(y) y^{\frac{1}{p}} + \mathbb{1}_{(1; \infty)}(y)$$
$$f_Y(y) = \frac{d}{dy} F_Y(y) = \mathbb{1}_{[0; 1]}(y) \frac{1}{p} y^{\frac{1}{p}-1}$$

UNIFORM DISTRIBUTION 2

$X \sim \text{unif}(1, 3), \quad Y = \sqrt[3]{\frac{5-X}{2}}$
$$f_X(x) = \mathbb{1}_{(1; 3)}(x) \frac{1}{2}$$
$$X = 5 - 2Y^3$$
$$g(y) = 5 - 2y^3$$
$$f_Y(y) = -f_X(g(y)) g'(y) \quad \text{monot. decr.}$$
$$= 3y^2 \mathbb{1}_{(1; 3)}(5 - 2y^3)$$
$$1 < 5 - 2y^3 < 3$$
$$\Leftrightarrow 1 < y < \sqrt[3]{2}$$
$$\Rightarrow f_Y(y) = 3y^2 \mathbb{1}_{(1; \sqrt[3]{2})}(y)$$

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